Mirroring

InterSystems

Creative data technology

Overview

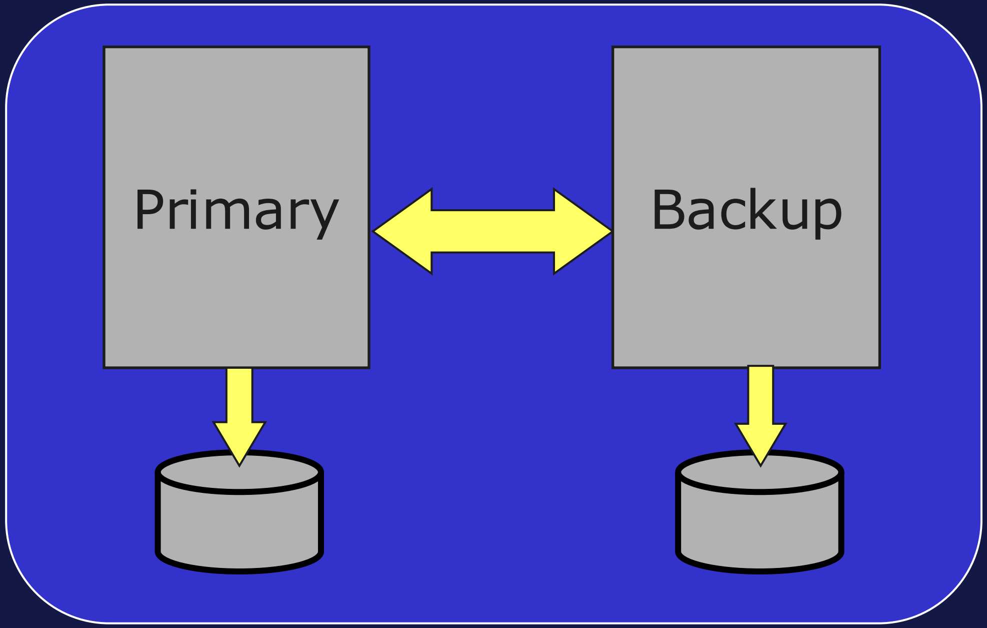
- Mirroring uses.
- Mirroring architecture.
- Async mirror members.
- Synchronization and mirroring.
- Failover.
- Configuration considerations.

Database Mirroring

- Economic, automatic failover option for high-availability.
- Provides solution for planned and unplanned downtime.
  - Minimizes shared-resource related outages with separate physical resources.
  - Uses logical data replication to reduce disk corruption risks.
  - Provides disaster recovery and business intelligence benefits with async mirror members.

Failover Member Set

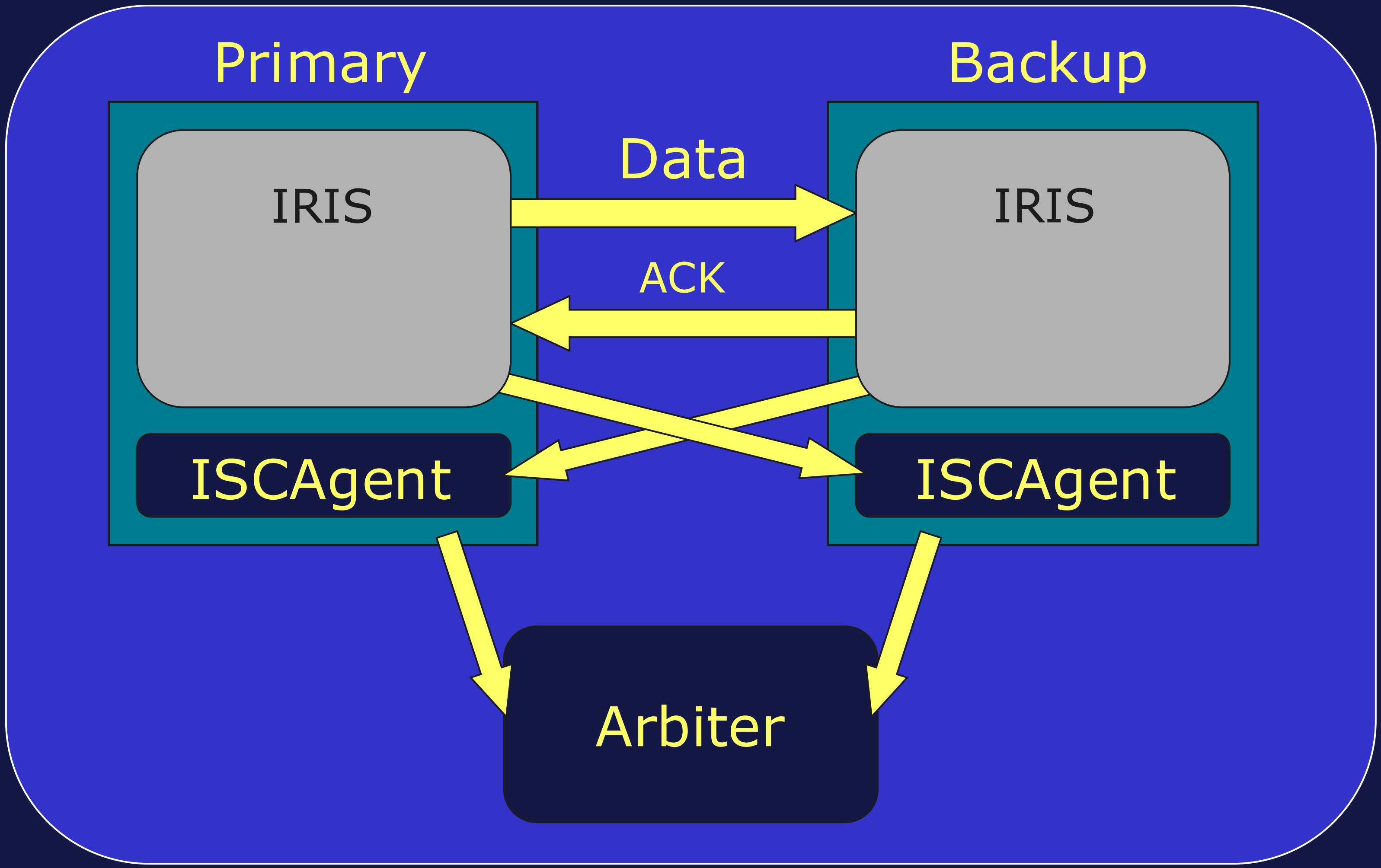
- Logical grouping of two IRIS instances connected via TCP/IP.
  - One system automatically elected primary.
  - Other becomes backup.



Mirror Channels of Communication

- Journal records flow from primary to backup.
  - **Not** database blocks.
- ACKs flow from backup to primary.
- Each failover member communicates with ISCAgent (OS-level process) on other member.
- Each failover member communicates with Arbiter (another ISCAgent) on separate server.

Mirror Channels of Communication (cont.)



ISCAgent

- OS-level process only.
- Uses port 2188.
  - Refer to documentation to change port number.
- Not started by default but required for mirroring.
- Reports status of mirror members.
- Can send journal files as needed to mirror members.
- Used to determine if automatic failover possible in event of system failure.

Arbiter

- ISCAgent running on separate server.
  - Can use ISCAgent of:
    - Any other instance.
    - Independent installation.
  - One Arbiter may serve many mirror sets.
  - Uses minimal resources.
    - Can operate on machine hosting other systems.
- Used to determine if automatic failover possible in event of entire primary server failure.
  - System crash, network isolation, and so on.

Mirroring, Databases and Journaling

- Mirroring done per database.
- All mirrored databases journaled.
- Mirrored databases read-only on backup.
- Primary pushes journal records to backup.
- When instance becomes primary:
  - Switches journal to mirror journal file.
    - Name starts with "MIRROR_".
    - Contains some failover member information.

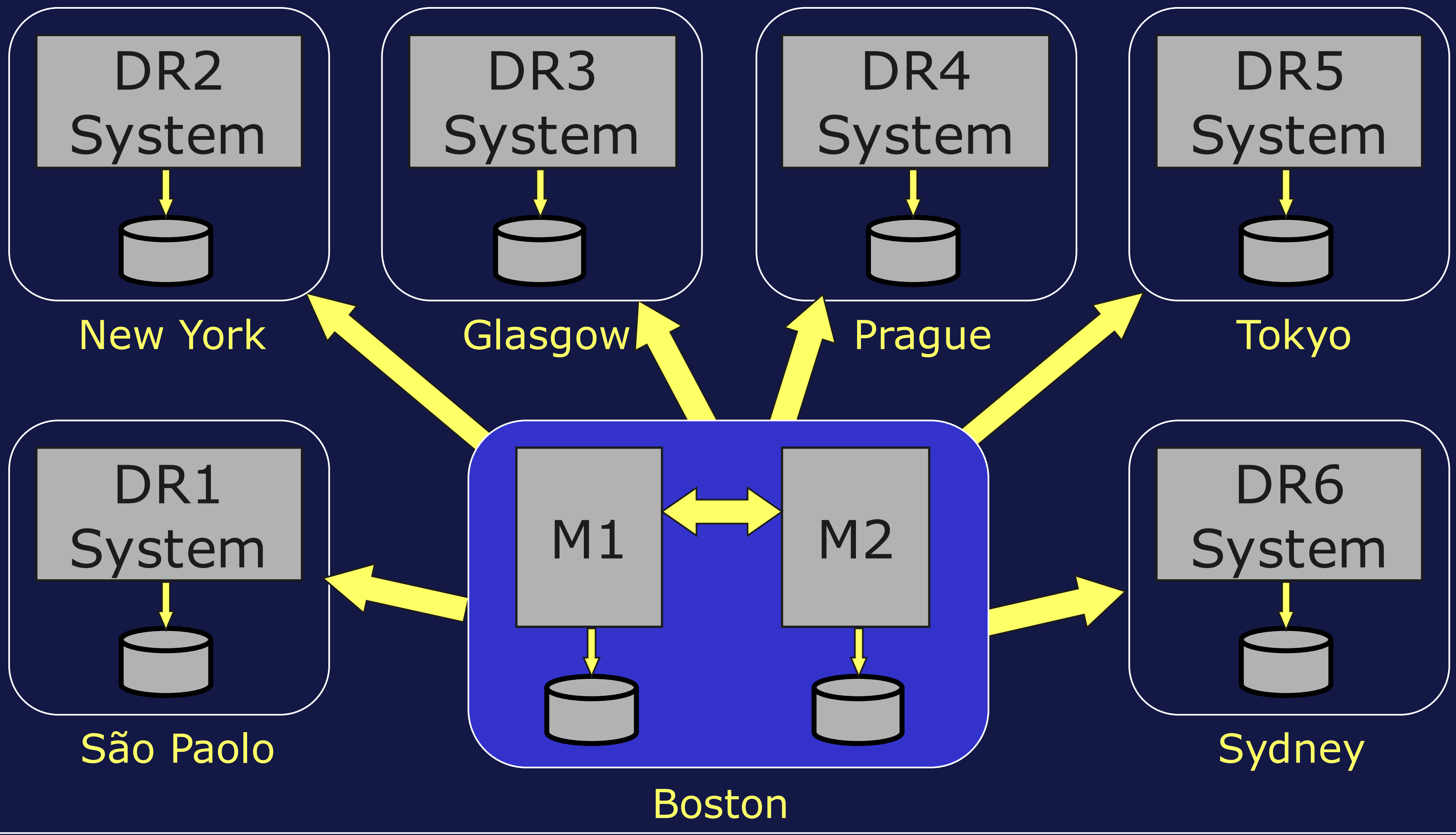
Asynchronous Mirror Members

- One async member can receive updates from several mirror sets for data replication.
- One mirror set can update up to 14 async members.
- Async mirror members:
  - Not candidates for automatic failover.
  - Get updates same as failover member.
  - Does not report its status to primary.
  - Can be used for Disaster Recovery, Business Continuity, Business Intelligence.

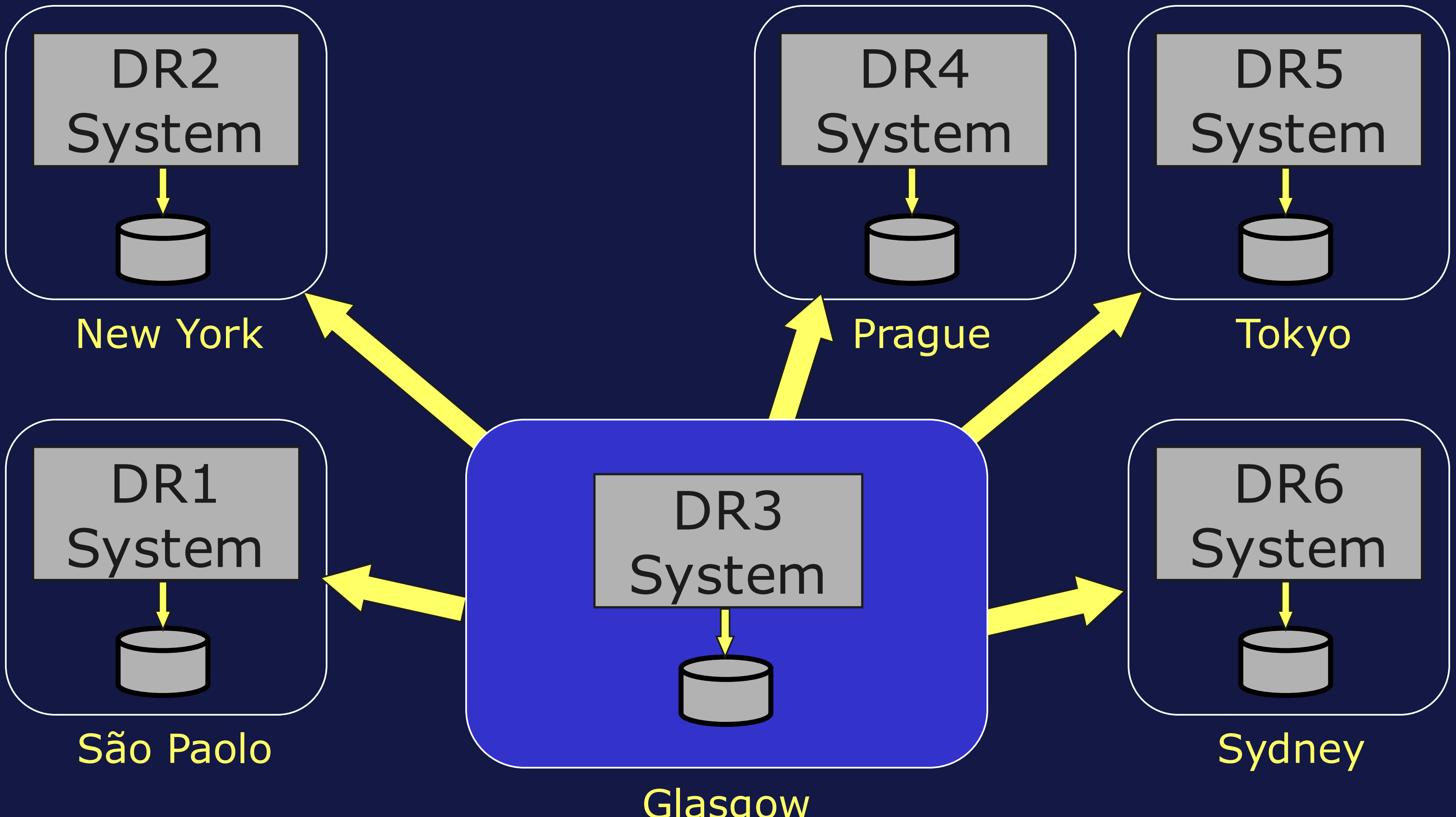
Async Types

- Disaster Recovery
  - Belongs to only one mirror set.
  - Read-only.
  - Can manually promote to failover member.
- Reporting
  - Can belong to multiple mirror sets.
  - Maintains either read-only or read-write copies of data.
  - Cannot be failover member.

Async Use Case: Disaster Recovery



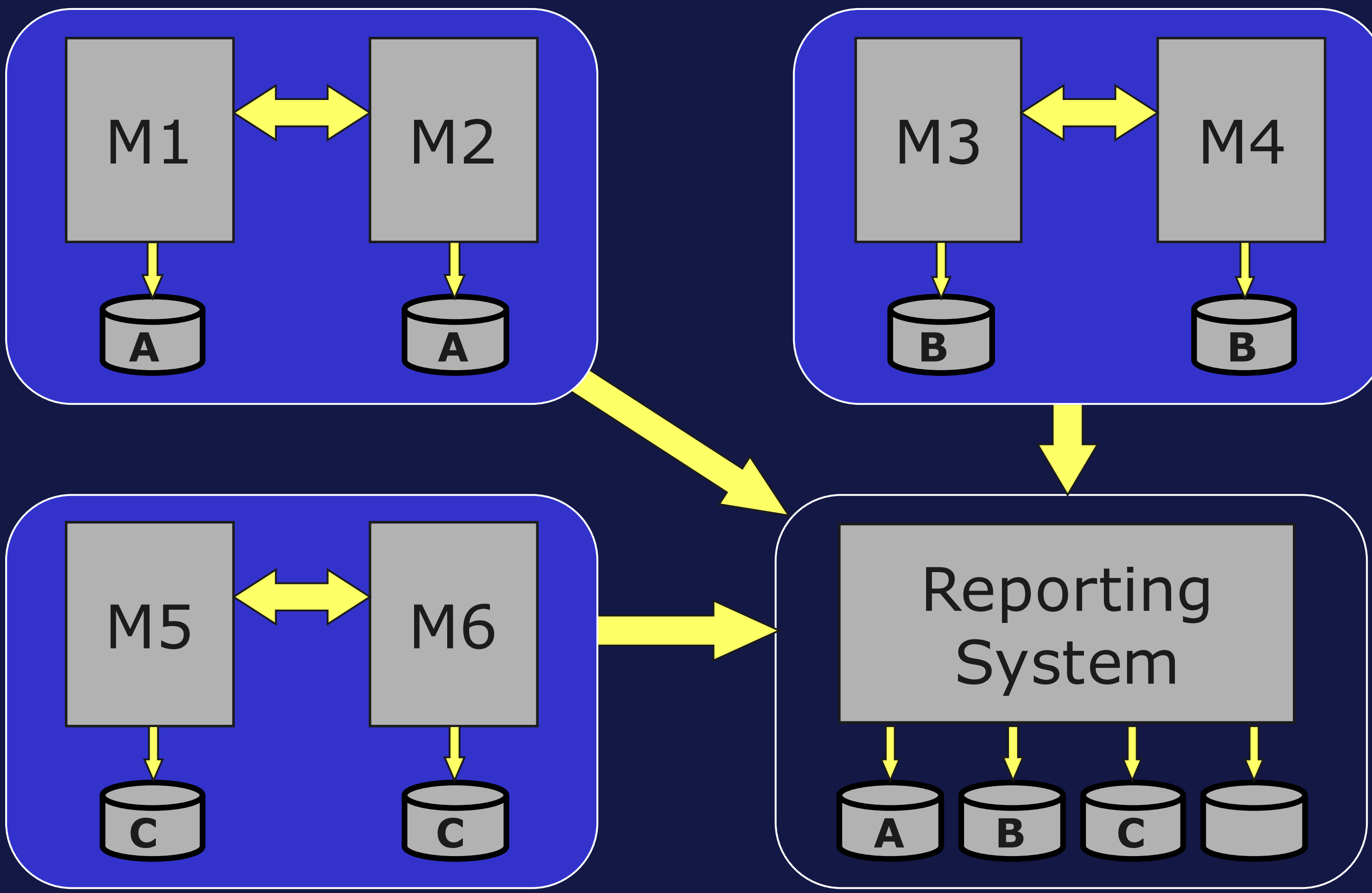
Async Use Case: Disaster Recovery (cont.)



Manually Promoting Disaster Recovery Async Members

- Can promote to backup.
  - Useful if one failover member will be down for extended period of time.
  - Obtains most recent journal data and becomes backup.
- Can promote straight to primary if both failover members are down.
  - Possible loss of application data.
- For additional options refer to *High Availability Guide > Mirroring > Configuring Mirroring.*

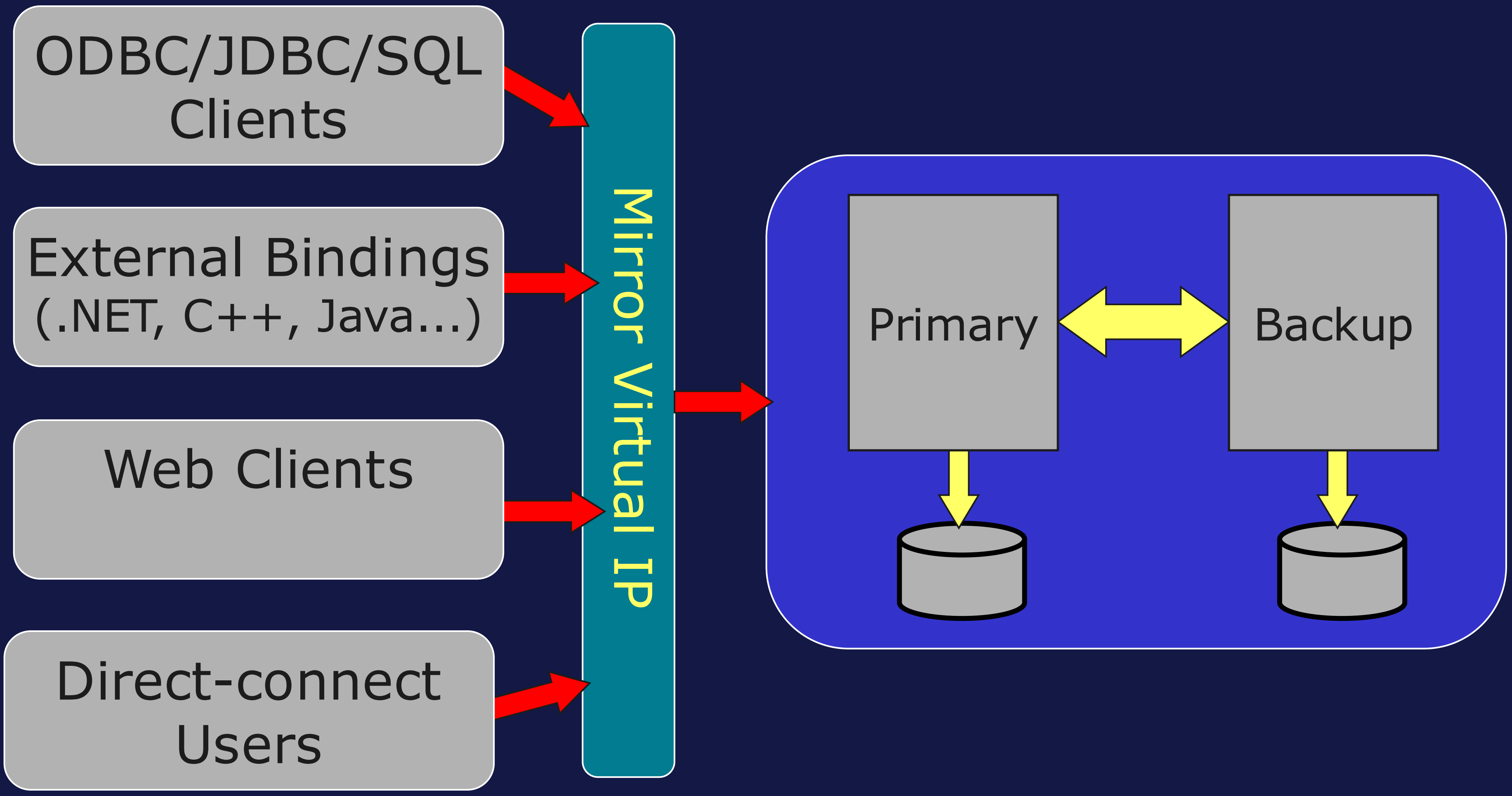
Async Use Case: Reporting System



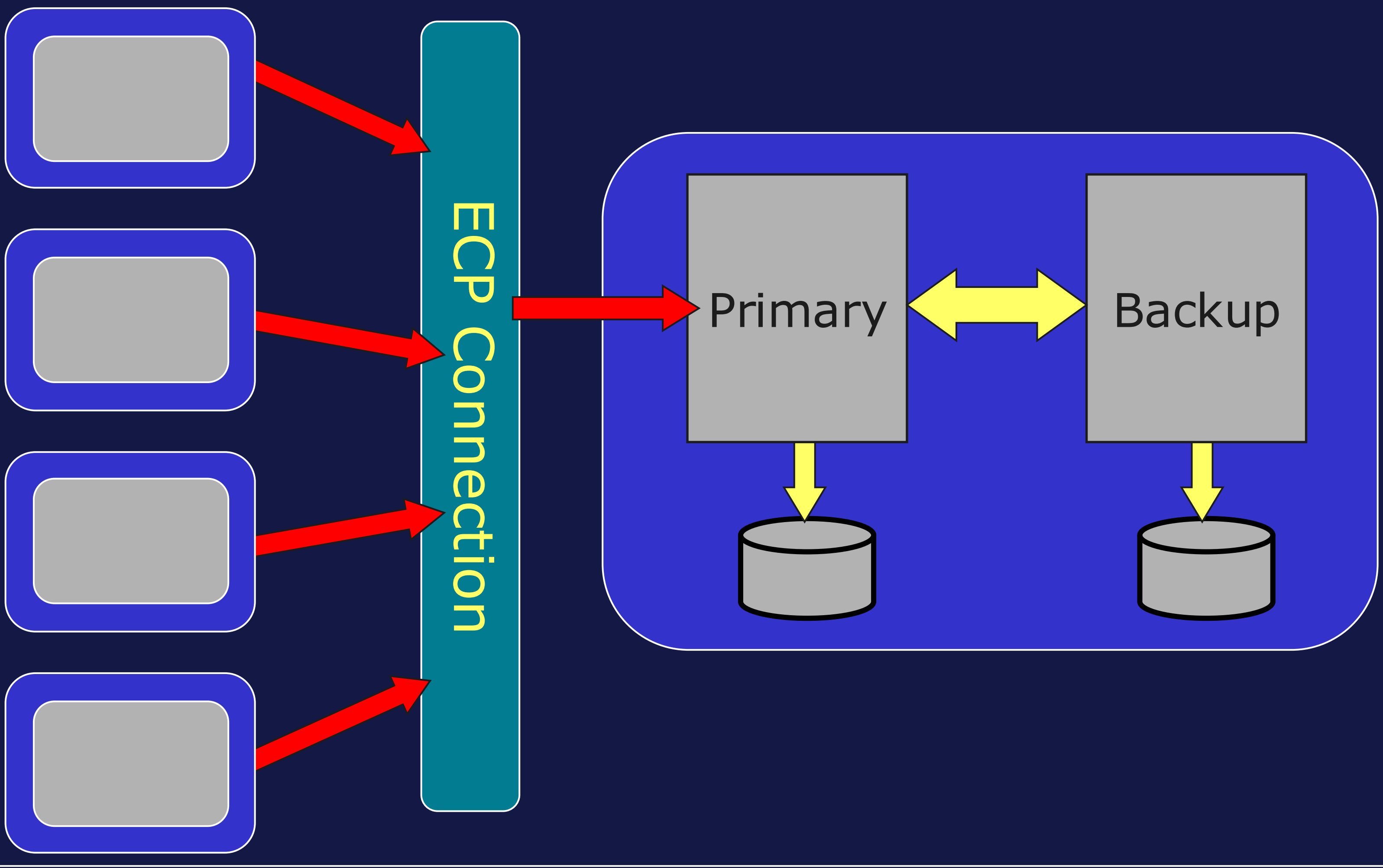
Applications and Mirroring

- All external clients connect via Virtual IP.
  - Includes language bindings, SQL clients, Management Portal, VS Code and others.
  - Use of Virtual IP recommended.
- ECP clients connect to primary.
- Applications view failover as server restart.
  - External clients:
    - Disconnect and re-establish connection to Virtual IP.
  - ECP clients:
    - Pause and automatically reconnect to current primary.

Virtual IP Connections



Application Servers Connecting to Mirrored Data Server



Backup States

- Backup member has two states.
  - Active.
    - Has all journal data needed to automatically failover.
    - Receiving journal records from primary member.
  - Catchup.
    - Does **not** have all journal data needed to automatically failover.
    - Receiving journal records from ISCAgent.
- Quality of Service (QoS) parameter used to determine if backup current.
  - Switched to Catchup if QoS timeout expires before acknowledgement returned from backup.

Mirror Failover Triggers

- Failover trigger indicates need to determine if failover possible.
  - If failover possible, backup becomes primary.
- Failover triggers include:
  - Backup does not hear from primary every QoS timeout period.
    - Network issues, system failure, and so on.
  - Data channel closed by primary.
    - System crash or hang on primary.
  - Manual failover.
    - ^MIRROR routine.
    - SYS.Mirror class.

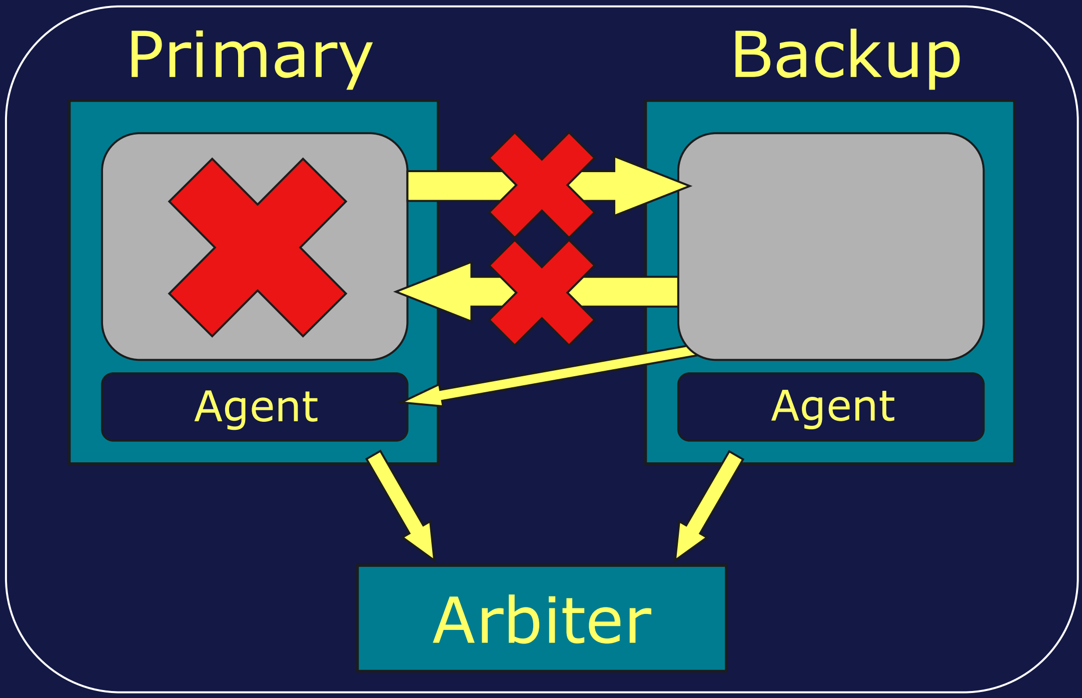
Determining Failover Possibility

- Before attempting automatic failover, must verify:
  - Primary really down or unreachable.
    - Not just lost connection between backup and primary.
- Otherwise:
  - Cannot perform automatic failover.

Determining Failover Possibility (cont.)

- Arbiter available.
  - Arbiter provides information to help backup determine if failover should occur.
    - Backup automatically takes over if both backup and Arbiter lose connection to primary.
    - Primary enters troubled state if disconnected from backup and Arbiter.
- Arbiter unavailable or not configured.
  - ISCAgent of primary provides information to help backup determine if failover should occur.

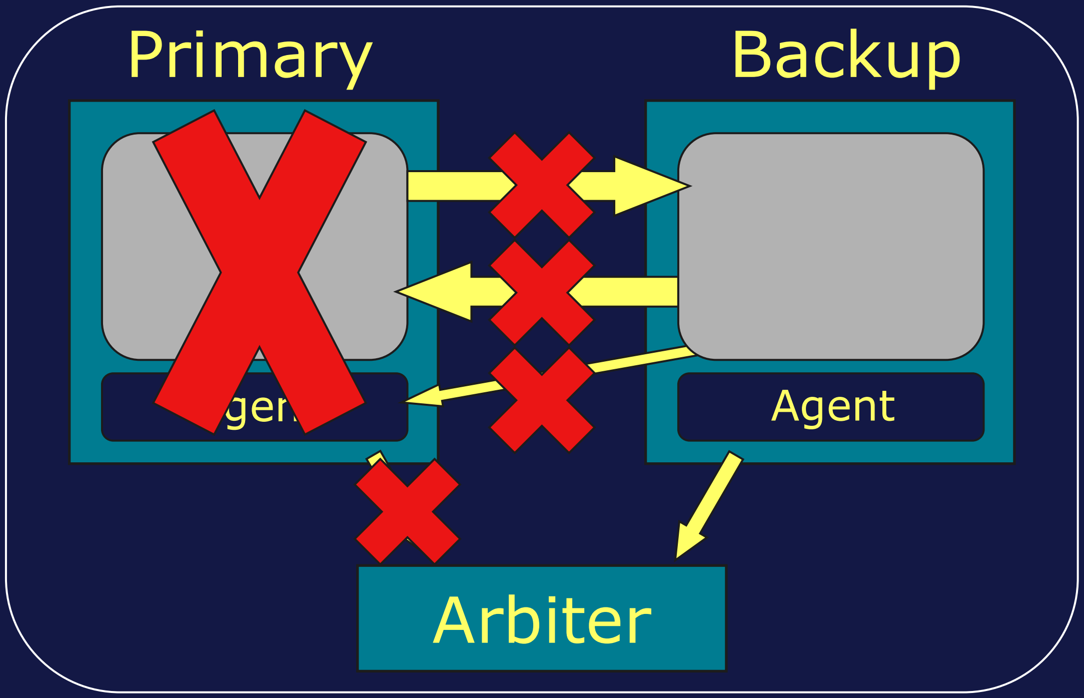
Failover Scenario 1: IRIS Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Running.
IRIS on Backup	Running.
Arbiter	Running.

- Arbiter confirms that primary is down.
- Automatic failover successful.

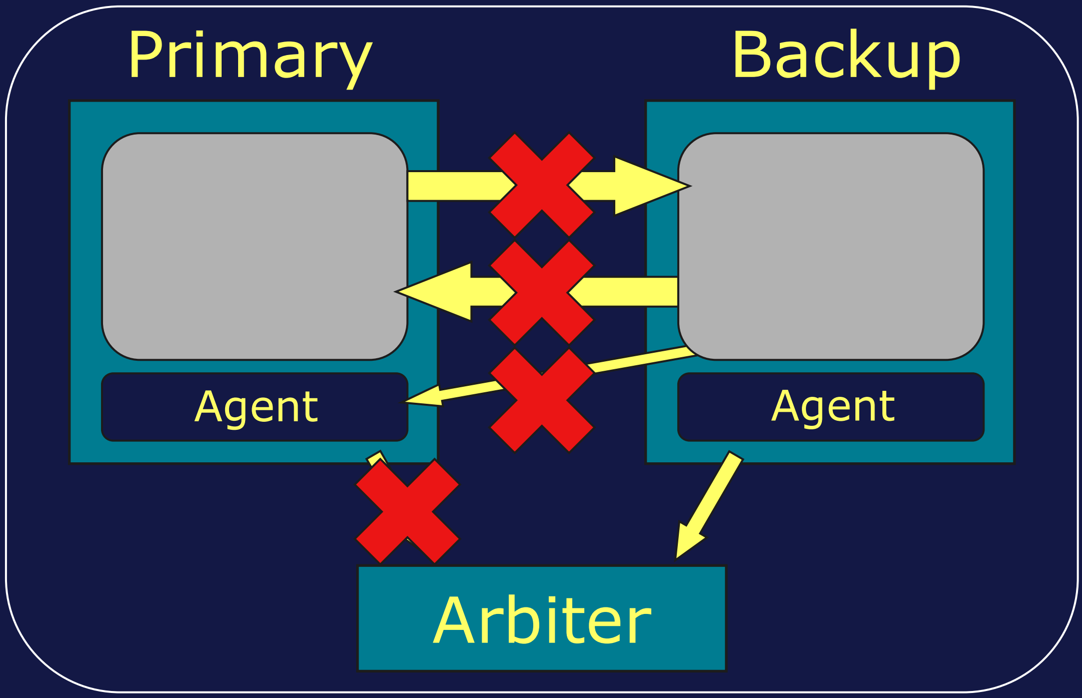
Failover Scenario 2: Primary Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Failed.
IRIS on Backup	Running.
Arbiter	Running.

- Arbiter confirms that primary is unreachable.
- Automatic failover successful.

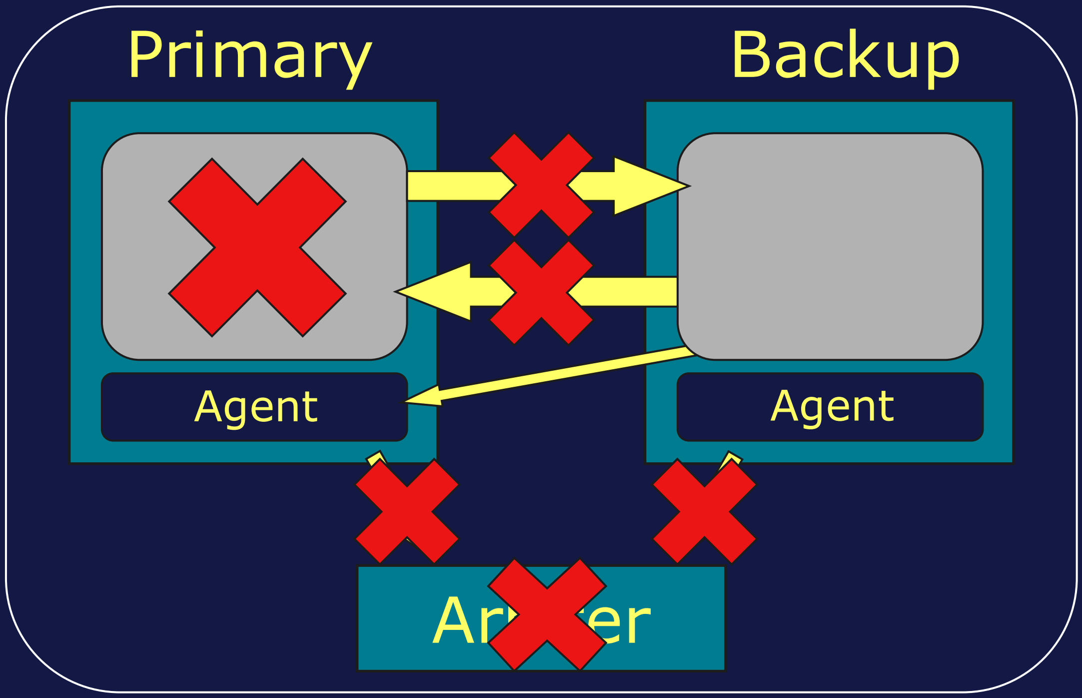
Failover Scenario 3: Primary Unreachable



Component/ System	Status
IRIS on Primary	Running.
ISCAgent on Primary	Running.
IRIS on Backup	Running.
Arbiter	Running.

- Arbiter confirms that primary is unreachable.
- Automatic failover successful.
  - Primary detects disconnection and enters troubled state.
    - Once primary reconnects, backup forces primary down.

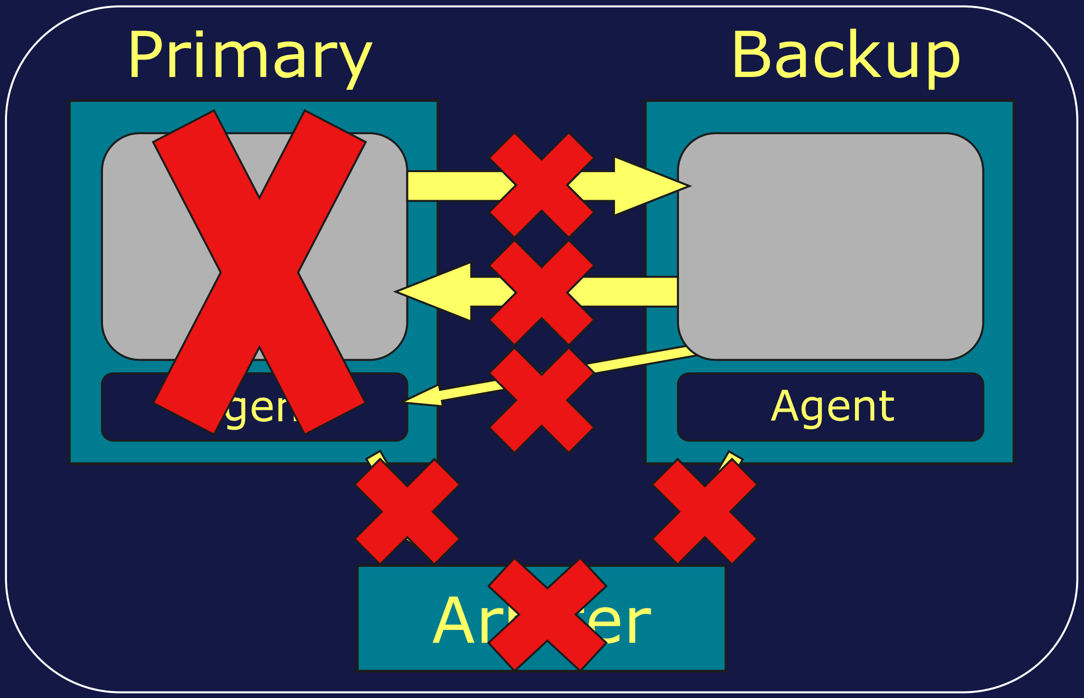
Failover Scenario 4: IRIS and Arbiter Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Running.
IRIS on Backup	Running.
Arbiter	Failed.

- ISCAgent on primary confirms that primary is down.
- Automatic failover successful.

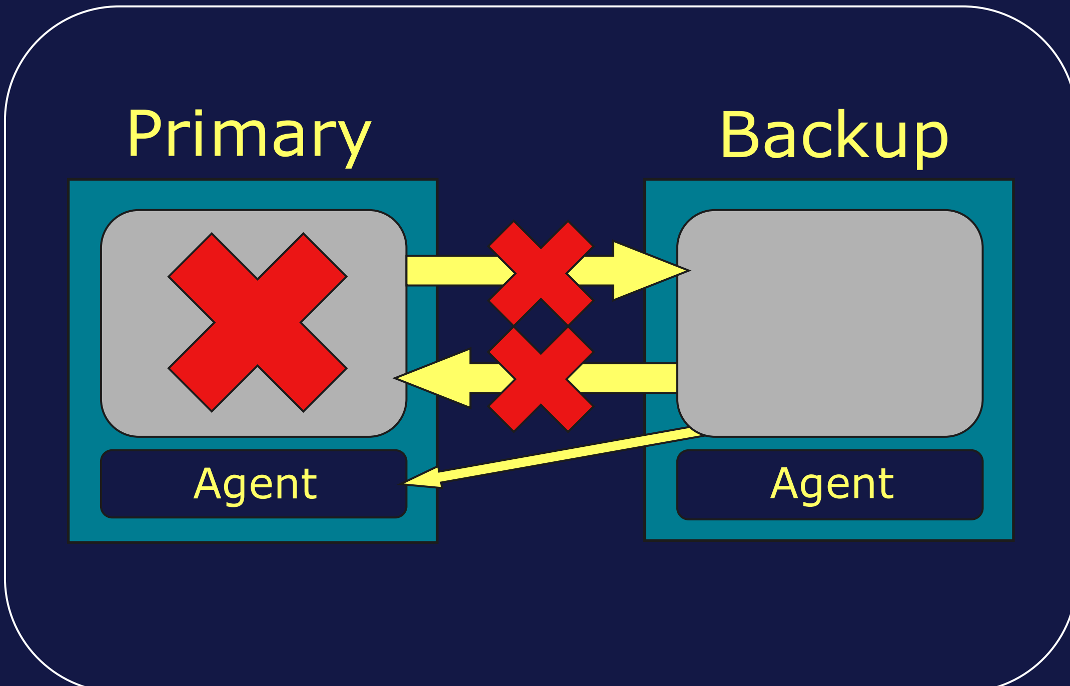
Failover Scenario 5: Primary and Arbiter Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Failed.
IRIS on Backup	Running.
Arbiter	Failed.

- Backup cannot determine status of primary.
- Automatic failover not possible.
  - May manually failover.

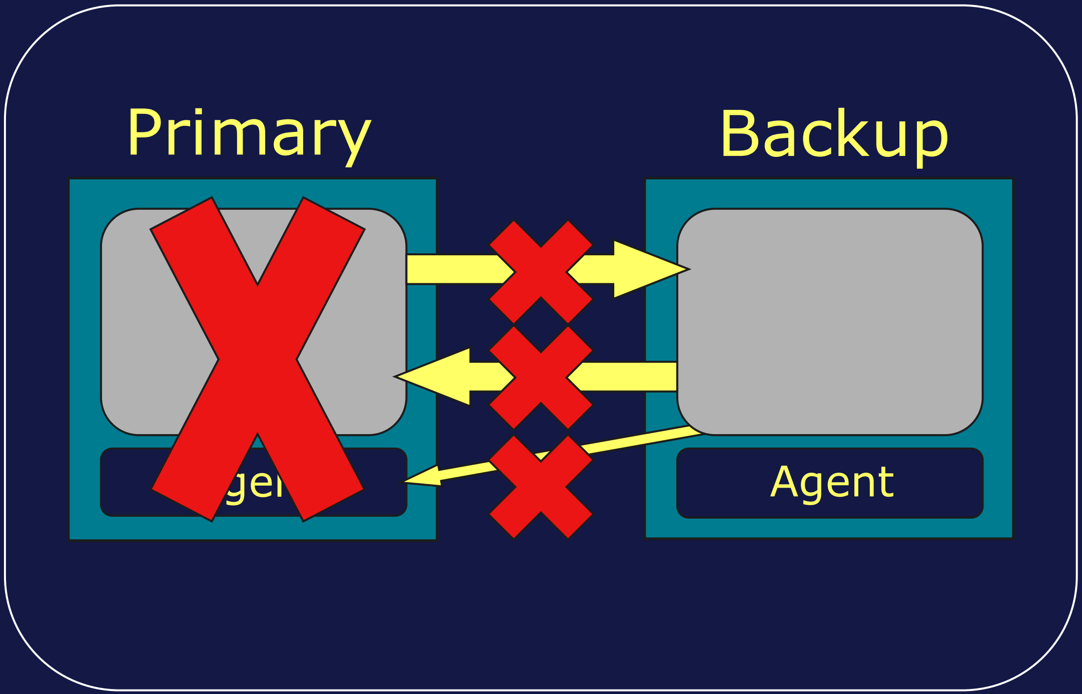
Failover Scenario 1 Without Arbiter: IRIS Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Running.
IRIS on Backup	Running.

- ISCAgent on primary confirms that primary is down.
- Automatic failover successful.

Failover Scenario 2 Without Arbiter: Primary Down



Component/ System	Status
IRIS on Primary	Failed.
ISCAgent on Primary	Failed.
IRIS on Backup	Running.

- Backup cannot determine status of primary.
- Automatic failover not possible.
  - May manually failover.

Configuration Considerations

- Configure Mirroring using Management Portal or ^MIRROR routine.
- Can be different:
  - Platform, OS and endian.
  - Product version.
- Both must have same character width.
- Must synchronize configuration details.
  - For example, namespaces, databases, global mappings, locales, and so on.

Network Considerations

- Network quality between failover members impacts:
  - Ability of backup to keep up with primary.
  - Failover time.
  - Performance of primary.
- High bandwidth, low latency, minimal loss network preferred.
  - Private network between failover members ideal.
- Set QoS timeout to at least double the round-trip time between failover members.

Additional Considerations

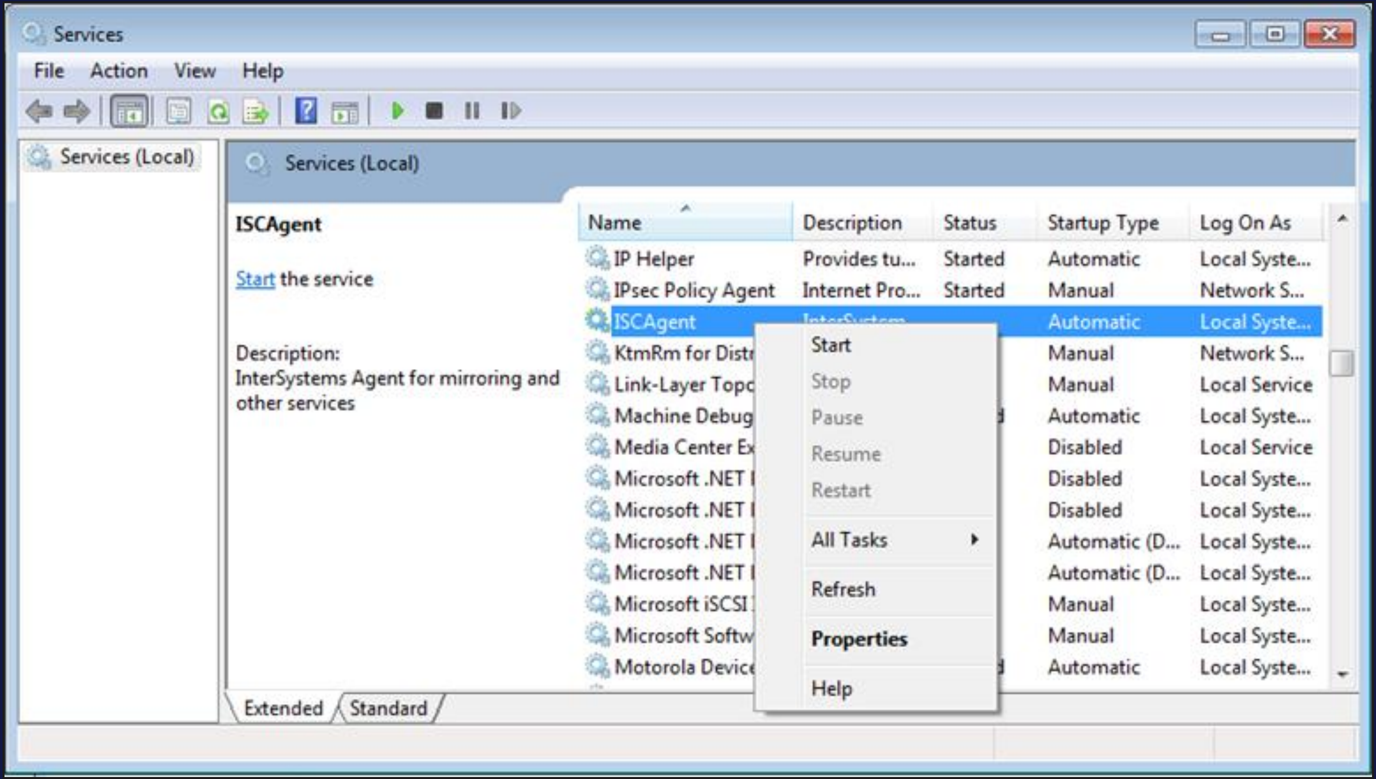
- Mirroring does not start in Emergency Access Mode.
- Database encryption must use same key on both systems.
  - Async members may use own key.
  - Must use TLS if using database encryption.
- DataCheck utility.
  - Can check consistency of data between failover members.
  - Refer to *Data Integrity Guide > Data Consistency on Multiple Machines* for details.

How To: Configure Mirror

- Steps:
  1. Create mirror and configure first failover member.
  2. Configure second failover member.
  3. If using TLS, add second failover member to first failover member.
  4. Configure async members. (optional)
- Additionally, on each member:
  - ISCAgent service must be enabled.
  - %Service_Mirror must be enabled.
  - Create databases, namespaces, users.

How To: Enable ISCAgent on Windows

- Navigate to services menu.
  - Start > Control Panel > Administrative Tools.
- Start ISCAgent service.

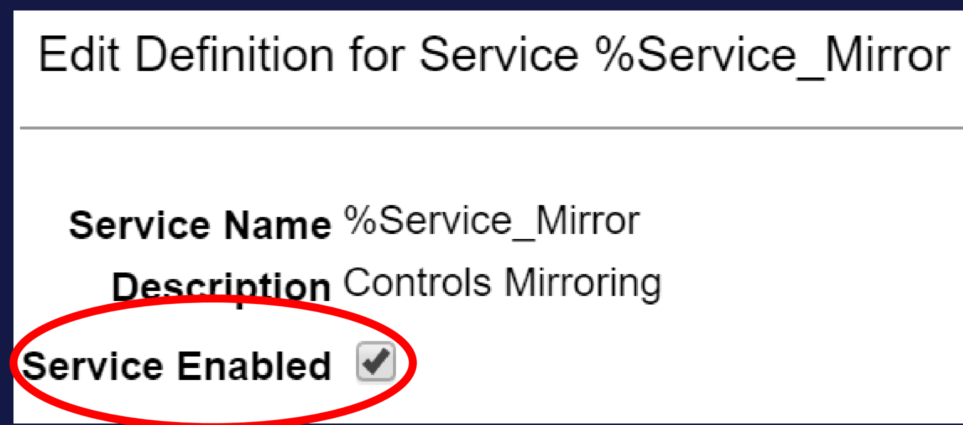


How To: Enable ISCAgent on Other Platforms

- Run ISCAgent start/stop script located in the following locations:
  - AIX: /etc/rc.d/init.d/ISCAgent
  - Linux: /etc/init.d/ISCAgent
  - MacOS: /Library/StartupItems/ISCAgent/ISCAgent

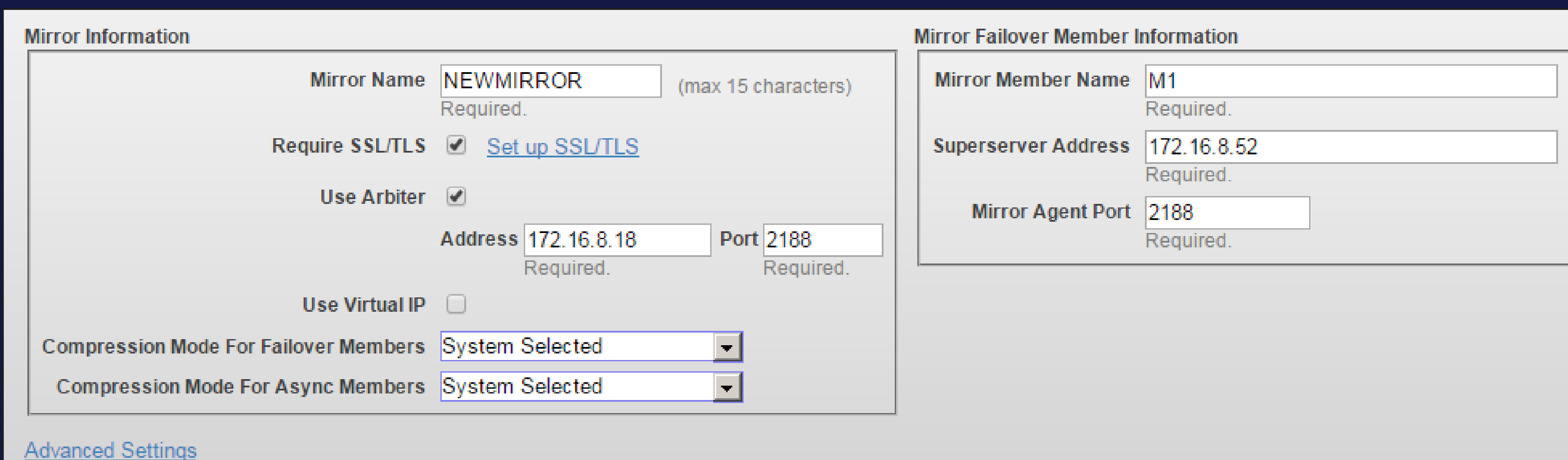
How To: Enable Mirror Service

- To enable %Service_Mirror, either:
  - System Administration > Security > Services.
  - System Administration > Mirror Settings > Enable Mirror Service

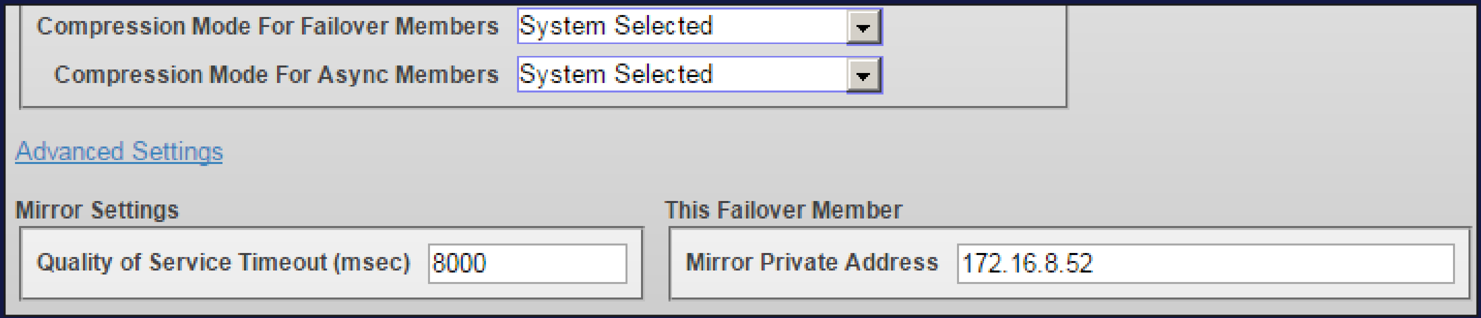


How To: Create Mirror and First Failover Member

- Create Mirror.
  - System Administration > Configuration > Mirror Settings.
  - Configure Name, TLS, Arbiter and Virtual IP Settings.
  - Member Name, Superserver Address and Mirror Agent Port automatically populated.



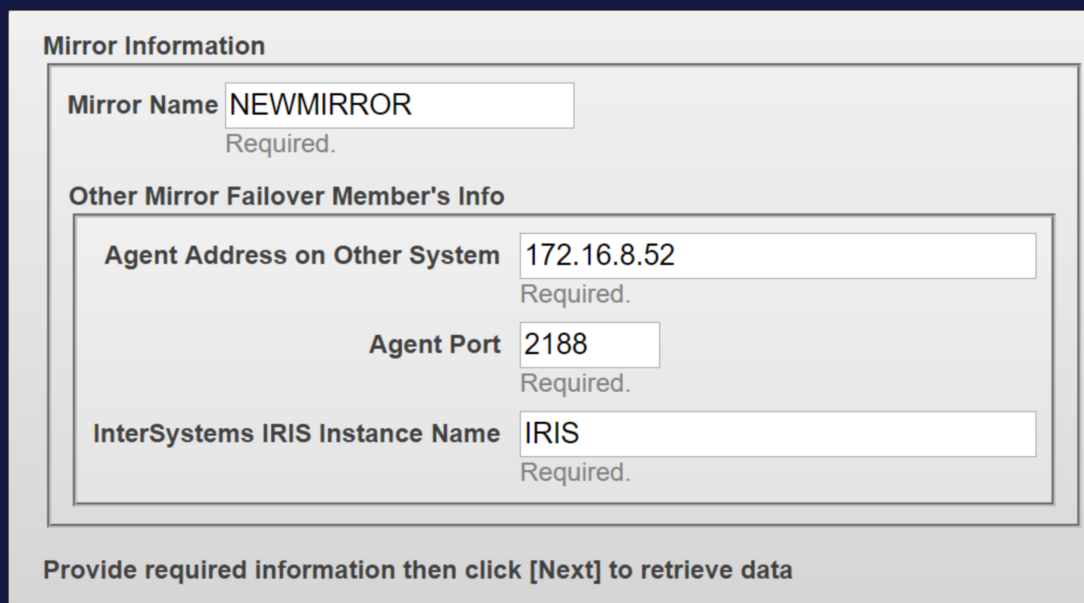
How To: Create Mirror and First Failover Member (cont.)



- Choose Compression Mode for data.
  - Default System Selected suitable for most environments.
    - Optimizes based upon network utilization.
    - May instead choose to always compress or never compress.
- Configure Advanced Settings. (optional)
  - Quality of Service Timeout:
    - Maximum time to wait for data to be acknowledged before demoting backup to catchup state.
  - Can set Private Address for Data, ACK and Agent channel communication.

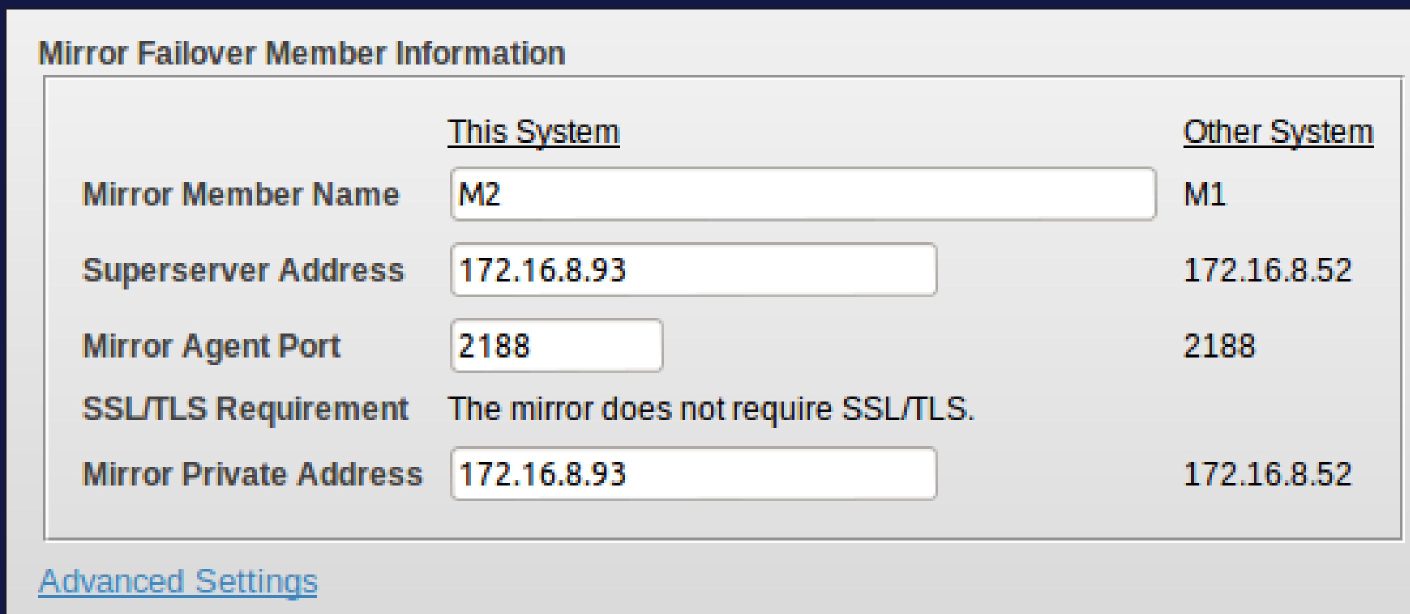
How To: Configure Second Failover Member

- System Administration > Configuration > Mirror Settings > Join as Failover.
- On second machine:
  - Fill in mirror settings.
  - Click Next.



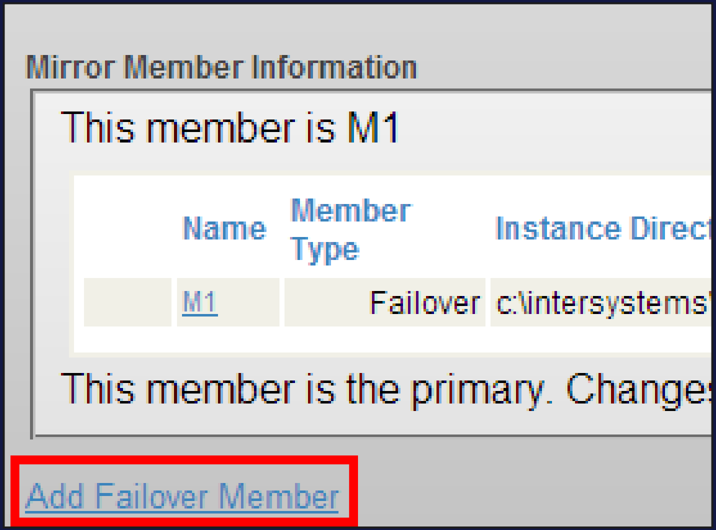
How To: Configure Second Failover Member (cont.)

- After connecting to mirror:
  - Edit settings for second member. (optional)
  - Click Save.



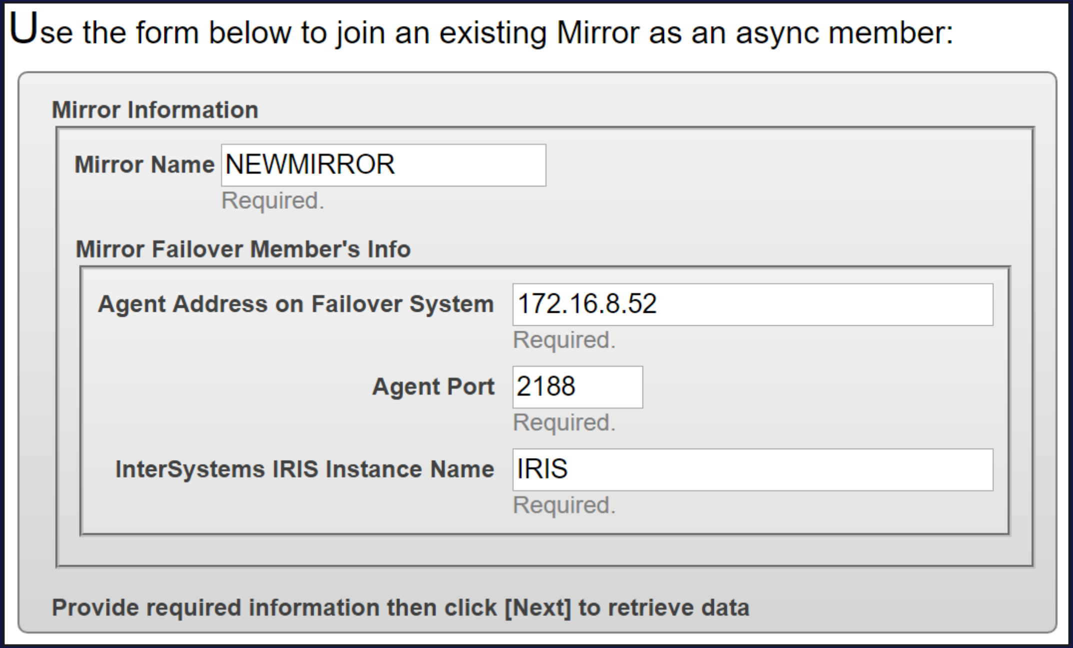
How To: Add Second Failover Member to First Failover Member

- If using TLS, must add second failover member to first failover member.
- On first machine:
  - Click Edit Mirror.
  - Click Add Failover Member.
    - Fill in IP, port and instance name of second machine.



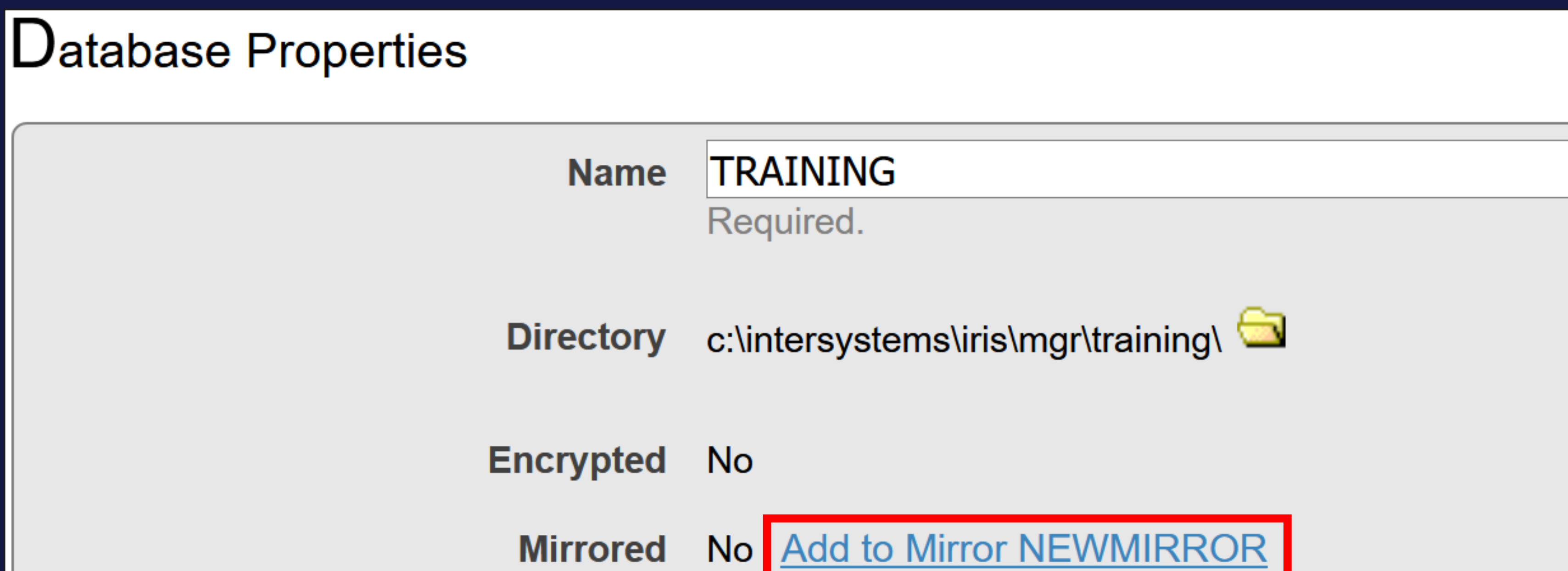
How To: Configure Async Members

- On other machines:
  - Click Join as Async.
  - Fill in mirror information and click Next.
  - Add async member information and click Save.



How To: Add Databases to Mirror

- To mirror databases:
  - Click Add to Mirror for mirrored databases on primary.
  - Backup databases from primary.
  - Restore databases to backup.



How To: Add Databases to Mirror (cont.)

- Instead of using backup/restore, mirror member can automatically download and transfer mirrored databases. (v2025.1+)
  - **Creation** of new mirrored database(s) on backup or async member triggers download and transfer from primary.
  - If databases are **not** large, easier alternative to manual backup and restore.

Suggested Readings

- *High Availability Guide.*

Summary

- What are the key points for this module?